Lab 08

Species classification with support vector machines

Teaching Week 08 | Individual work. Submit in R or Python.

Purpose

A support vector machine draws the widest street between the classes, and the street depends on the units of the predictors, on the fine C and on the reach γ. You will classify fish families or tree-seedling species from their measurements, show what scaling changes, tune C and γ inside cross-validation, try class weights, and compare the tuned SVM with kNN or LDA on the same folds. The lab opens on October 29, 2026, with the class on support vector machines, and runs for one week.

Learning outcomes

After completing this lab, you should be able to

- explain why an SVM boundary depends on the predictors' units, and keep the scaling inside the pipeline;
- tune C and γ inside cross-validation on a powers-of-ten grid, and read the grid;
- judge class weights with balanced accuracy and the recall of each class; and
- compare a tuned SVM with kNN or LDA fold by fold on the same outer folds.

Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

Option A | Beaufort Sea Marine Fishes Project (BSMFP) 2013 - Fish identification and measurements

Publisher. Fisheries and Oceans Canada
Official record. https://open.canada.ca/data/en/dataset/74e3dbd2-1845-11ef-90aa-8b219c568296
Record identifier. `74e3dbd2-1845-11ef-90aa-8b219c568296`
Licence and format. Open Government Licence – Canada; CSV with a PDF data dictionary
Expected variables. station; latitude; longitude; date; gear; family; species; weight; total, fork and standard length; liver and gonad weight; sex; maturity
Use in this lab. classify a fish's family from its measurements
Why it fits. About 10 000 fish from 75 trawl stations, sampled in August and September 2013, in 13 named families plus a few unidentified fish, which you drop. Two thirds are cod (Gadidae),

so merge the rare families into one class by a stated rule and judge every model by balanced accuracy. Weight, total length and standard length are recorded for almost every fish. Fork length is recorded for only three families, and liver weight, gonad weight, sex and maturity are missing for most fish in a pattern that follows the family, so whether they are present already gives the family away. Latitude and longitude mark the station, though 11 of the 75 stations have none. Fish from one station share conditions, so the station is a natural group.

Option B | FunRoots: Root and leaf traits of tree seedlings of seven species in Eastern Canadian Forests

Publisher. Natural Resources Canada
Official record. https://open.canada.ca/data/en/dataset/d9c4d93f-2388-4158-b3cf-49a9ecea6b2c
Record identifier. `d9c4d93f-2388-4158-b3cf-49a9ecea6b2c`
Licence and format. Open Government Licence – Canada; CSV tables with a data dictionary and a protocol
Expected variables. seedling; region; light condition; plot; species; clade; sampling date; age; height; stem diameter; crown; leaf and root traits; biomass
Use in this lab. classify a seedling's species from its size, leaf and root traits
Why it fits. The seedlings table holds 338 seedlings of seven species, 20 to 61 each, from 14 plots in three regions, under open or closed canopy. Several traits are sums or ratios of others (total biomass is above-ground plus below-ground biomass), and the clade follows from the species, so justify every predictor you keep. Petiole traits are recorded only for the broadleaf species and needle mass only for the two conifers, so their presence gives the species away. All seedlings in a plot share its light condition and its site, so the plot is a natural group.

Data contract

Choose one listed dataset and build one multiclass modelling table.

- Define one target with at least three classes, the fish family or the seedling species. Merge or drop rare classes by one stated rule, and report the class counts before and after.
- Retain at least 250 complete rows. For option A you may work on a stratified random sample of at least 2 000 fish, drawn with a recorded seed, to keep the tuning to minutes.
- Keep at least four numeric predictors with genuine variation, each justified from the dictionary. A predictor you derive from measured ones counts, such as the fish's condition, weight over length cubed, or the ratio of standard to total length. Identifiers, station and plot codes, latitude and longitude, the species when the target is the family, and any column whose presence or absence follows the target (fork length, liver and gonad weight, petiole and needle traits) are not predictors.
- Keep the station or plot of every row for the limitations, even though it is not a predictor.
- Fix five stratified outer folds once, with a recorded seed, and score every model on those same folds.
- Fit the scaling, and choose C, γ, k and the class weights, inside each outer training set only.
- Report balanced accuracy and the recall of every class, beside a baseline that always predicts the most common class.

- Document the official record URL, publisher, licence, retrieval steps, and access date.

What the contract does not settle

Meeting the contract does not show that the SVM is the better model, or that a score from random folds would hold at a station or plot the model has never seen.

Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I defined a target with at least three classes and stated the rule that merges or drops rare classes.
- ☐ My table satisfies every rule in the data contract.
- ☐ I kept identifiers, station and plot codes, and every column that restates the target or whose gaps follow it out of the predictors.
- ☐ I fixed five stratified outer folds with a recorded seed for every model to share.
- ☐ I put the scaling inside the pipeline, so it is fitted on training rows only.
- ☐ I will report balanced accuracy and the recall of every class.

Cumulative method

This lab carries forward the earlier data audit, cleaning, graphics, baselines, class-aware metrics, kNN and LDA, and nested cross-validation with the same folds for every model. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

Work program

1. Build the table and the folds

State the target, the merge rule with the class counts before and after, and each predictor with its unit and the reason it should carry information about the class. Report how many rows each gap removes, and decide what a gap means before you drop it. Assign every row to one of five stratified outer folds with a recorded seed and save the assignment. Score a baseline that always predicts the most common class on those folds. Its balanced accuracy is one over the number of classes, the floor every model must clear.

2. Scale inside the pipeline

Fix $C = 1$ and $\gamma = 1/p$, where p is the number of predictors, so that both languages fit the same model. Fit this radial SVM on the outer folds on the raw predictors, then again after rewriting one predictor in a smaller unit, such as weight in milligrams instead of grams, and report the outer balanced accuracy of both. Repeat the pair with standardization in front of the SVM. Explain from the street, which is measured in the units of the predictors, why one pair changes and the other does not. The scaler is

part of the model, so it is fitted on each training set and only applied to the rows it scores. In Python, put `StandardScaler` and `SVC` in one `Pipeline`. In R, `e1071::svm` standardizes inside each fit by default, so pass `scale = FALSE` for the raw fits.

3. Tune C and γ inside, score outside

Inside each outer training set, choose C and γ by five stratified inner folds on a grid in powers of ten, starting from C in 0.1, 1, 10, 100 and 1000 and γ in 0.001, 0.01, 0.1, 1 and 10. Refit the chosen pair on the whole outer training set and score it once on its outer fold. Show the inner grid of one outer fold as a table or heat map and read it. Say where the model is too stiff, where it is too flexible, and whether a diagonal of similar scores trades C against γ. If the best cell sits on an edge of the grid, widen the grid in that direction and search again. Report the pair each outer fold chose, and what it means when the folds disagree.

4. Try class weights

Balanced class weights give every class the same total weight. In scikit-learn pass `class_weight="balanced"`, and in R pass the same numbers as a named `class.weights` vector. Add the weighting to the inner search as a third setting, none or balanced, so that each outer training set chooses it together with C and γ, and report what each outer fold chose. Then set the weighted and unweighted SVMs side by side on the same outer folds by balanced accuracy, the recall of each class, and plain accuracy, with the confusion matrix summed over the outer folds for each. Say which classes gained, which lost, and what plain accuracy alone would have led you to conclude. These outer scores describe what the weights do, and the choice itself was already made inside.

5. Compare on the same folds and state the limitation

Fit kNN with k chosen inside each outer training set, or LDA, on the same predictors, scaling and outer folds. Report one row per outer fold with the balanced accuracy of the baseline, the comparison model and your tuned SVM. Because the folds are shared, the fair comparison is the difference on each fold, the SVM minus the comparison model. Report the mean and the standard deviation of those five differences, and say whether the mean difference is large against their spread. The corrected resampled t-test, with $1/4$ as the overlap term for five folds, turns that judgement into a p-value if you want one. Then name the natural groups as a limitation. Fish from one station, or seedlings from one plot, share conditions, and random folds put a test row's station or plot mates in training. Say in which direction this is likely to move every score in your table and what a prediction at a new station or plot would require. Grouped folds are not required.

Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Submit your own R or Python source and the evidence supporting the report. Include `folds.csv` with the outer fold of every row, so the marker can confirm that every model saw the same folds; `grid.csv` with the inner balanced accuracy of every C and γ pair for each outer fold, widened grids included;

and `fold-scores.csv` with the outer balanced accuracy and per-class recall of every model on every fold. Record the seeds, the merge rule, the retained rows, and any sample you drew.

Downloaded data and generated results are not course source, so they never go into the course repository. Keep the work and its evidence in your private work repository until the deadline has passed and the instructor opens sharing, then publish it in your public GitHub portfolio repository. Link rather than redistribute source data unless its licence and size permit it.

Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-08/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on November 5, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-08/`. A branch URL or movable tag alone is not a fixed submission. The package includes

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, figures, and any summary statistics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

Assessment standard

The course uses one rubric across all integrated labs. The best nine of the ten weekly lab results contribute 15% of the course grade.

Table 1. Integrated lab marking criteria.

Criterion	Weight (%)
Dataset provenance and suitability	15
Reuse of earlier methods	20
Current-week analytical method	35
Interpretation and limitations	20
Reproducibility and communication	10
Total	100

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.